The N=48 minimum does NOT transfer: the cut is load-bearing at N=96

N=96, degree 5, 4 colours, leak-free | job 51656823, 4 cells | s*_96 = 0.119, frozen at registration

THE TRANSFER FAILS, AND EVERY SIZE CLAIM HERE BECOMES AN N=48 CLAIM

stripped 0.850 vs control 0.676, +0.175 = 1.47 s*_96, OUTSIDE the registered 0.119 and in the direction that matters. The MLP, the width, or both are load-bearing at N=96. Every 'smallest architecture' claim in this programme is relabelled as an N=48 claim. This was a named outcome before the run closed and it is reported as a result, not as a disappointment.

THE CONTROL IS NOT ONE DRAW (F2 HELD)

Two N=96 control seeds existed at registration, median 0.621 over the bracket [0.564, 0.679]. The third seed was bought by this job precisely because a two-seed control was named as the most likely way stage F ends up weaker than it looks. It landed inside [0.564, 0.679] and the 3-seed median is 0.676 against the 2-seed 0.621, so the control's level is quotable.

WHAT IS DELIBERATELY NOT ON THESE SLIDES

No N=48 rho_50 shares an axis with an N=96 one -- the grids end at 0.33 and 0.32. No plateau-escape count is quoted: finding 44's 1.64 is an N=48 number and N=96's plateau has never been measured, so the training panel is drawn against a measured BAND instead. s* = 0.1484 is an N=48 number and is used nowhere here. The N=48 standing minimum (attention-only L=16 d192, 2,504,890 trainable at rho_50 0.580) does not move on anything in this job (F3).

What the model actually sees

Every token is one categorical id into a 34-symbol vocabulary (N=24, q=4, degree=5). No text, no digits-as-characters.

VOCABULARY

0-5	BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING
	BEGIN_ROUND SOLVED UNRESOLVED
	markers (UNRESOLVED = DD got stuck; unused here)
6-9	COLOR_0 COLOR_1 COLOR_2 COLOR_3
	the q colours (token name COLOR_n, prose 'colour')
10-33	N0 N1 ... N23
	one id per node, no structure in the id

GRAMMAR

BEGIN_HINTS	(NODE COLOR)*
BEGIN_GRAPH	(NODE (NODE){degree})*
BEGIN_REASONING	(BEGIN_ROUND (NODE COLOR)+)*
SOLVED	

*regex: * = zero or more, + = one or more, {degree} = exactly 5*

The graph section is a neighbour table, not an image: node, then its 5 neighbours.

THE ORDER WITHIN A ROW IS THE LEAK.

THIS INSTANCE, 199 TOKENS

HINTS	BEGIN_HINTS N0 COLOR_0 N3 COLOR_0 N5 COLOR_3 N6 COLOR_1 N7 COLOR_3 N8 COLOR_2 ... (31 tokens)
GRAPH	BEGIN_GRAPH N0 N1 N14 N9 N15 N17 N1 N5 N2 N14 N23 N0 ... (145 tokens)
REASONING	BEGIN_REASONING BEGIN_ROUND N2 COLOR_2 N12 COLOR_2 N15 COLOR_3 N18 COLOR_0 N19 COLOR_0 N21 ... (23 tokens)

TRAINING LOSS is next-token cross-entropy on EVERY token, the problem statement included -- there is no section mask.
AT EVAL the model is PROMPTED with everything through BEGIN_REASONING and generates the rest: 3 rounds, 9 deductions,
18 node+colour tokens, plus the BEGIN_ROUND markers and the SOLVED/UNRESOLVED terminator, which are trained targets like any other token.

DD is single-candidate elimination -- sudoku's 'naked single' rule transplanted to graph colouring, which is where the code's sudoku names come from; the task itself is the graph, not a 9x9 grid. Instance shown: seed 50000000009, unconstrained ensemble.

One real reasoning trace

15 hints of 24 nodes, then 3 rounds of simultaneous deduction. Hints: N0=COLOR_0 N3=COLOR_0 N5=COLOR_3 N6=COLOR_1 N7=COLOR_3 N8=COLOR_2 ...

THE TRACE, AS EMITTED

```
BEGIN_ROUND          N2 COLOR_2 N12 COLOR_2 N15 COLOR_3 N18 COLOR_0 ...
  6 deduced
BEGIN_ROUND          N1 COLOR_1 N11 COLOR_2
  2 deduced
BEGIN_ROUND          N4 COLOR_3
  1 deduced
SOLVED
```

WHY ONE DEDUCTION FIRES

node N2, neighbours N3 N19 N23 N17 N1

their colours known before this round:

```
   [0, 1, 3]  (3 = q-1 of the 4)

   (3 known neighbours, 3 DISTINCT colours)

   still unknown: N19 N1
```

only colour left: 2

```
so the trace emits  N2 COLOR_2
```

The model is never shown this reasoning -- only the node/colour pair.

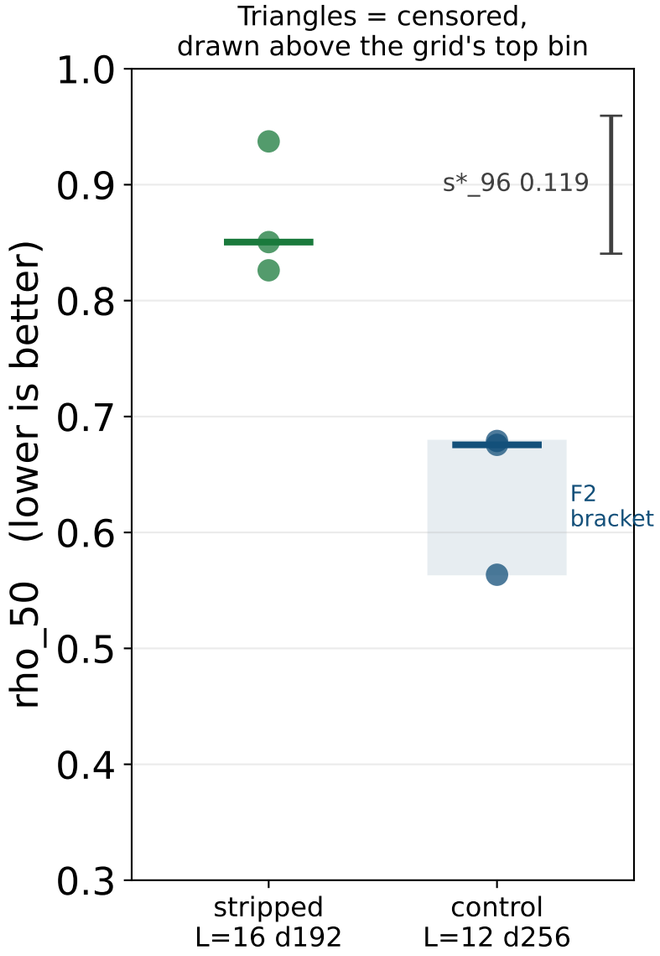
A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

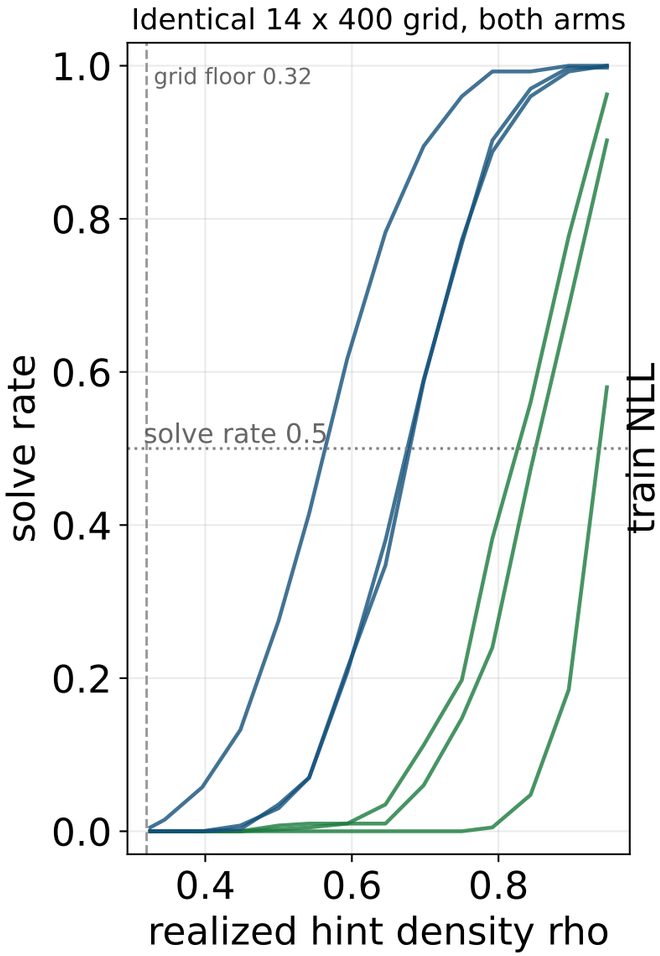
Drawn at N=24 so the adjacency fits on a slide. The scheme is identical at N=96, where the run lives; only the sequence gets longer.

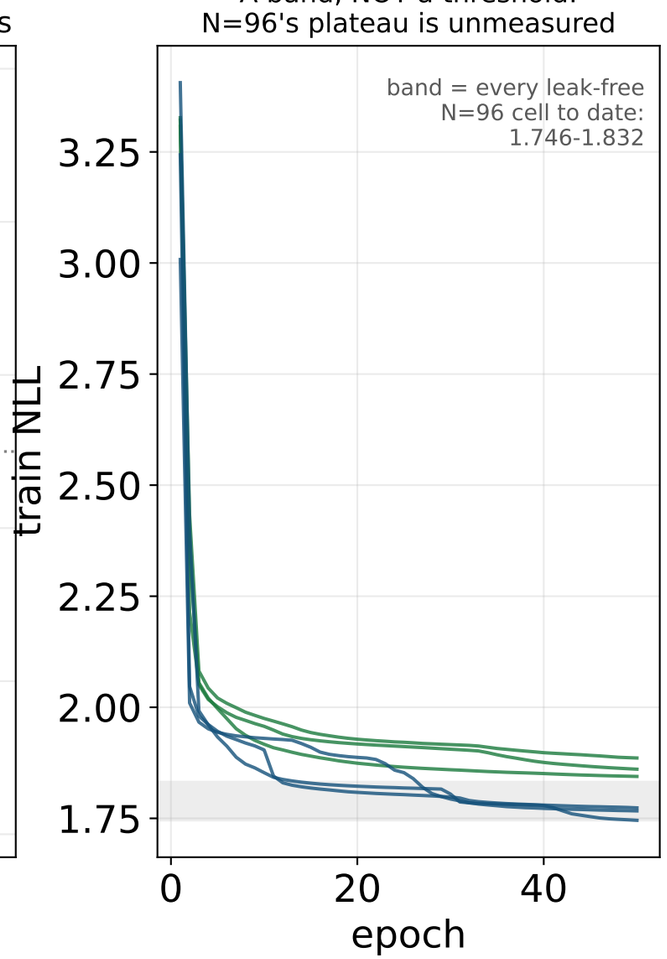
T=3 rounds, the median of 40 draws from this config (range 1-10) -- one instance, shown to make the task legible; no result in this deck is computed from it. The worked deduction on the right uses the state BEFORE this round, which is what DD itself saw; it is recomputed from the graph, not read out of the model.

The transfer test at N=96, arm against paired control

Left: every seed, not just the median. Middle: the curves rho_50 is read off. Right: training, against the MEASURED N=96 band







arm	rho_50	mean	sprd	cross
stripped	0.850	0.181	0.160	3/3
control	0.676	0.423	0.162	3/3

arm	trainable	total
stripped	2,618,602	8,929,514
control	9,790,058	9,790,058

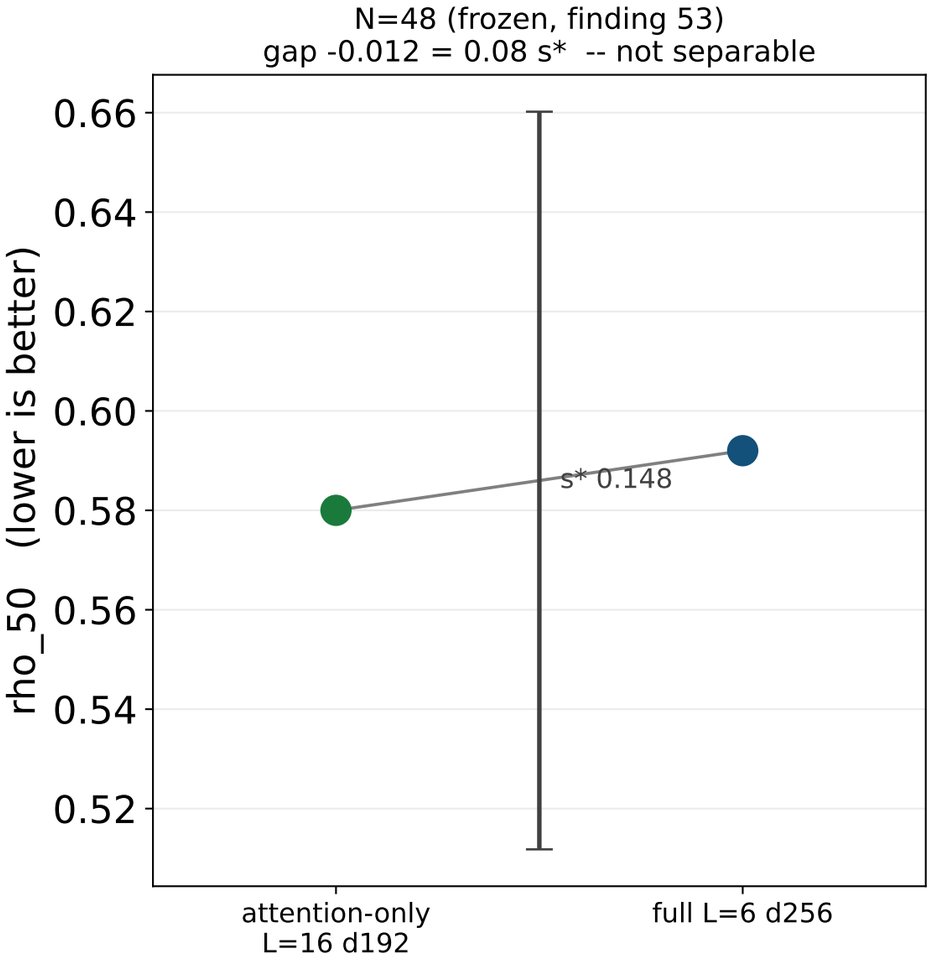
gap +0.175 = 1.47 s*_96

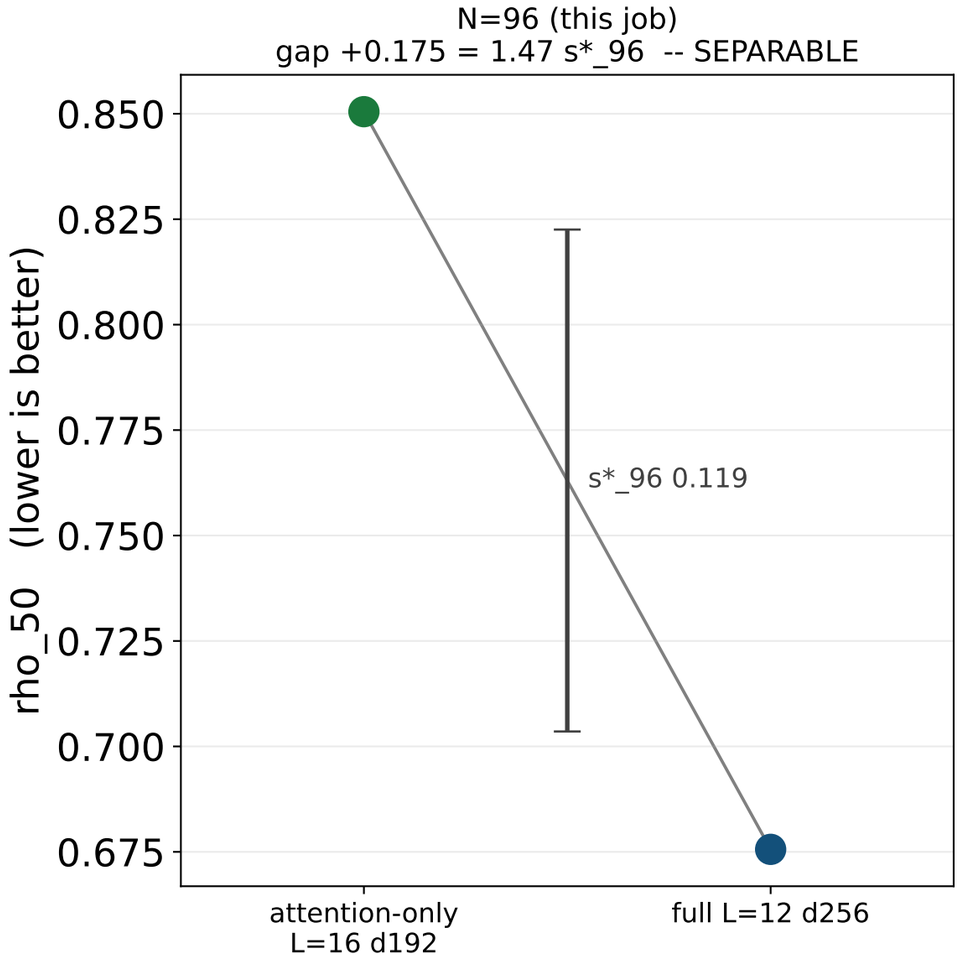
stripped attention-only L=16 d192 control full L=12 d256

spin_gnn @ fb2029c | job 51656823 | scorer scripts/stagef_score.py, registered before the first cell landed | N=96, 14 x 400 leak-free (--shuffle-adj --fall), 3 seeds

Two problem sizes, two yardsticks, deliberately not one axis

Each panel is a gap against ITS OWN control, in ITS OWN s*. The grids end at 0.33 and 0.32; the numbers are not comparable





The left panel does not move on anything in this job (F3). F can corroborate or undermine the N=48 minimum; it can never replace it.

The registered predictions, scored

Named in scripts/stagef_score.py before the first F cell had an eval on disk

F1 IT TRAINS AT ALL -- HELD

Registered: 3 of 3 stripped seeds produce a MEASURED rho_50 on a grid whose floor is 0.32. Observed: 3 of 3 crossing. F1 gates F0, because an unmeasured rho_50 cannot be compared to a level, and it gates the READING of a censor: three censors is an architecture statement, one or two is finding 51's seed story.

F0 THE HEADLINE -- HIGHER

Registered: |stripped - control| <= s*_96 = 0.119, frozen from the three two-seed N=96 rungs that existed at registration (L=12 fall 0.115, L=8 fall 0.138, L=12 cosine 0.104). Observed: stripped 0.850 vs control 0.676, +0.175 = 1.47 s*_96. BOTH falsification directions were given a reading in advance, which is what stops either one from being re-narrated after the fact.

F2 THE CONTROL IS NOT ONE DRAW -- HELD

Registered: the third control seed lands inside [0.564, 0.679] AND the 3-seed median stays within s*_96 of the 2-seed 0.621. Observed: s0 0.679, s1 0.564, s2 0.676, median 0.676. If F2 falsifies, F0 survives only as an ORDERING and the levels are not quotable.

F3 THE N=48 STANDING MINIMUM DOES NOT MOVE ON N=96 EVIDENCE

A REGISTRATION, not a measurement. The standing minimum is attention-only L=16 d192, 2,504,890 trainable at rho_50 0.580, an N=48 ladder result under rule H3 = I3 = J3, on a grid whose floor is 0.33. Stage F can CORROBORATE or UNDERMINE it; it can never replace it. Written down so that no number on these slides can later be read as having moved it.

THE YARDSTICK, RE-ESTIMATED BUT NEVER SUBSTITUTED

s*_96 = 0.119 is what every verdict above uses, and it was frozen before these cells ran. The scorer re-estimates it from whatever is on disk now and prints both; if the two disagree, the registered one still stands and the disagreement is itself the thing to report.

What stage F leaves open

Written before the run closed, so these are not the questions the answer happened to leave

THE RELABELLING IS THE WORK

Every 'smallest architecture' claim in this programme becomes an N=48 claim, including the ladder's own ~2M floor. The next question is WHICH cut is load-bearing at N=96 -- the MLP or the width -- and that is a two-cell experiment against this same control, not a sweep.

THE N=48 LADDER IS NOT CLOSED BY THIS

Stage K (51668331) tested whether the ~2M floor measured on the L=12 width ladder belongs to the task or only to L=12, at L=16 d160 and d128. Its verdict is on its own slides, in N=48's own yardstick; nothing here moves it either way, and nothing there moves anything here.

THE OPEN N=48 QUESTION, STILL DELIBERATELY UNBUNDLED

The width curve is convex on [192, 256] at L=12: all three registered interpolations overestimated rho_50 at d224 (J1). A fourth width point is the only way to know the shape of that stretch. It is worth 3 cells and it was queued behind this job, because F decides whether any N=48 number is worth generalizing in the first place.

WHAT A READER SHOULD CHECK FIRST

The per-seed levels, not the medians. The seed spread at a fixed architecture has exceeded the depth gaps throughout this programme, and s*_96 = 0.119 is a MEASURED spread, not a confidence interval. If a gap on these slides is smaller than that bar, the reading is 'no separable difference' -- and no page here says otherwise.